Lesson 13: Integral Test

Key Assessment Criteria Rubrics

- **Function Definition and Validity:** Correct identification of $f(x)$ from the series a_n.
- **Condition Checking** (25%):
  - **Positive:** Verify $f(x) > 0$ for all $x \geq N$.
  - **Continuous:** Verify $f(x)$ has no gaps or asymptotes on $[N, \infty)$.
  - **Decreasing:** Verify $f(x)$ is non-increasing. A common technique is to show the derivative $f'(x) \leq 0$ for $x \geq N$.
- **Improper Integral Evaluation** (50%):
  - Express the integral as a limit: $\lim_{t\to\infty} \int_1^t f(x)dx$.
  - Use techniques like substitution or integration by parts correctly.
- **Interpretation of Result** (25%):
  - If $\int_1^\infty f(x)dx < \infty$ (finite value), the series **converges**.
  - If $\int_1^\infty f(x)dx = \infty$ (or diverges), the series **diverges**.
  - **Important:** The value of the integral is *not* the sum of the series.